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Calcium Imaging Correlation and Synchrony Metrics Guide

This guide explains which metrics to use for different types of calcium imaging data, with visual examples showing what each metric measures.

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1. $\Delta F/F$ Calcium Traces (Raw Fluorescence)

Recommended Metric: Zero-lag Pearson Correlation

What it measures

Zero-lag Pearson correlation measures **slow co-fluctuations** between neurons: - Whether two ROIs tend to increase or decrease together - Similarity of $\Delta F/F$ trace shapes over time - **NOT timing or causality** — only covariation

Why this metric?

Calcium traces are inherently slow (hundreds of milliseconds to seconds), representing the integrated activity of neurons. Because of this temporal blurring, precise timing information is lost. Correlation is the only meaningful measure for continuous, slow-varying signals.

Visual Example

Interpretation

- **High positive correlation ($r = 0.8-1.0$):** Both traces consistently rise and fall together (see ROI 1 vs ROI 2 above)

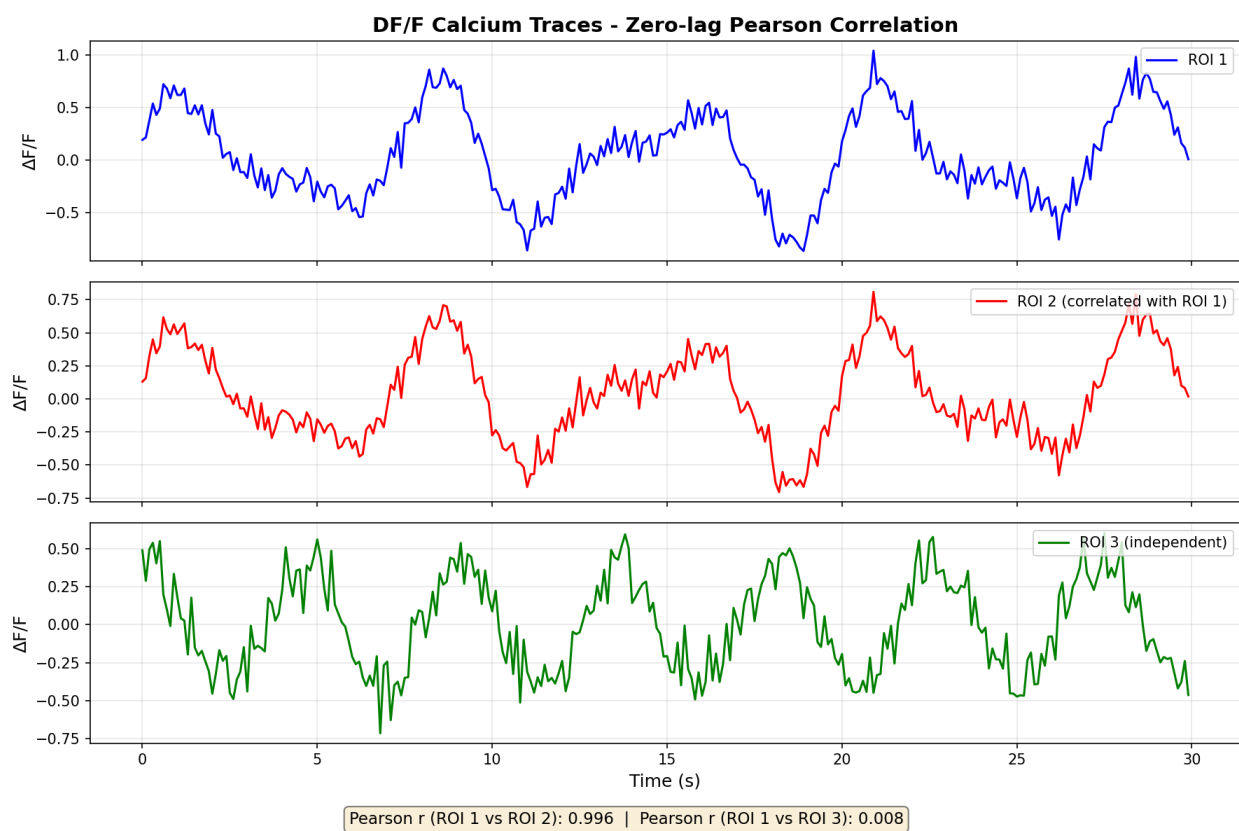


Figure 1: DF/F Traces Correlation

- **Near-zero correlation ($r \approx 0$):** Traces fluctuate independently (see ROI 1 vs ROI 3 above)
- **Negative correlation ($r < 0$):** When one trace rises, the other tends to fall

When to use

Use for: - Measuring ensemble coactivity between neurons - Identifying functional connectivity at the population level - Comparing global activity patterns across brain regions

Avoid: - Trying to infer precise timing relationships - Measuring synchrony (use event-based data instead) - Detecting causality or directionality

2. Deconvolved $\Delta F/F$ Traces (OASIS $C(t)$)

Recommended Metric: Zero-lag Pearson Correlation

What it measures

$C(t)$ represents a **denoised, sharpened version** of the raw $\Delta F/F$ trace: - Removes slow baseline drift - Sharpens calcium transients - Still a continuous signal, **not discrete events**

Why this metric?

While deconvolution improves signal quality and temporal resolution, $C(t)$ remains a continuous trace. It does **not** restore spike-level timing precision. Therefore, correlation is still the appropriate metric.

Visual Example

Interpretation

- **High correlation:** ROIs show similar patterns of transient activity
- **Low/zero correlation:** Independent dynamics
- The correlation is often **cleaner** than raw $\Delta F/F$ because noise is reduced

When to use

Use for: - Improved correlation estimates compared to raw $\Delta F/F$ - Analyzing shared transient dynamics after noise removal - Better signal-to-noise ratio for functional connectivity analysis

Avoid: - Synchrony metrics (timing is still imprecise) - Assuming spike-level temporal precision

3. Calcium Peaks (Event Detection)

Calcium peaks represent **approximate event times** detected from $\Delta F/F$ or $C(t)$ traces. They have inherent temporal jitter of ~100-300 ms, so only lag-tolerant metrics are appropriate.

Recommended Metrics

A. Event Cross-Correlation (with lag tolerance)

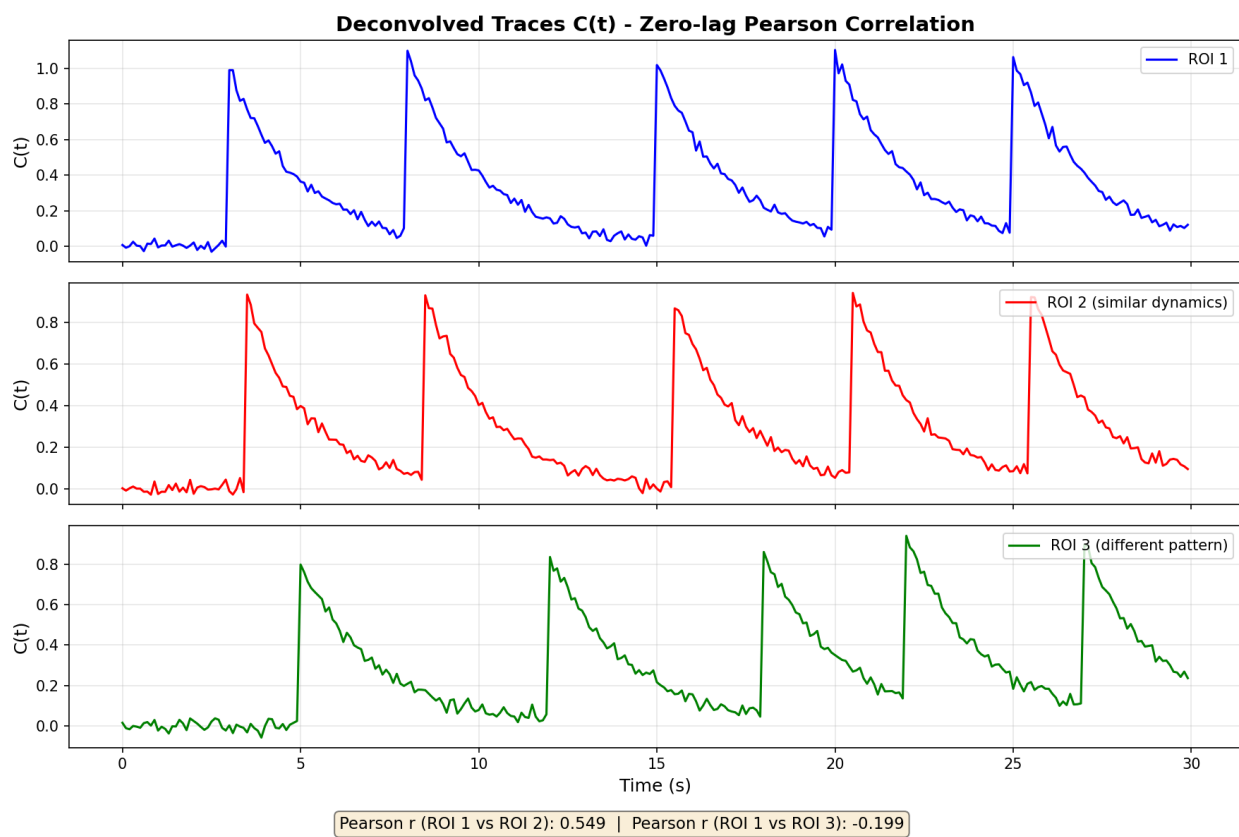


Figure 2: Deconvolved $C(t)$ Traces

What it measures

Whether peaks from two ROIs tend to occur **near each other in time**, allowing for temporal jitter:
- Rough temporal lag relationships - Strength of co-activation - Which ROI tends to activate first (coarse temporal ordering)

B. Jitter Synchrony ($\pm$ window)

What it measures

Whether two ROIs have **significantly more co-occurring events than expected by chance**:
- Allows a time tolerance window (e.g., ± 200 ms) - Returns a synchrony score between 0 and 1 - Robust to temporal jitter inherent in calcium imaging

Visual Example

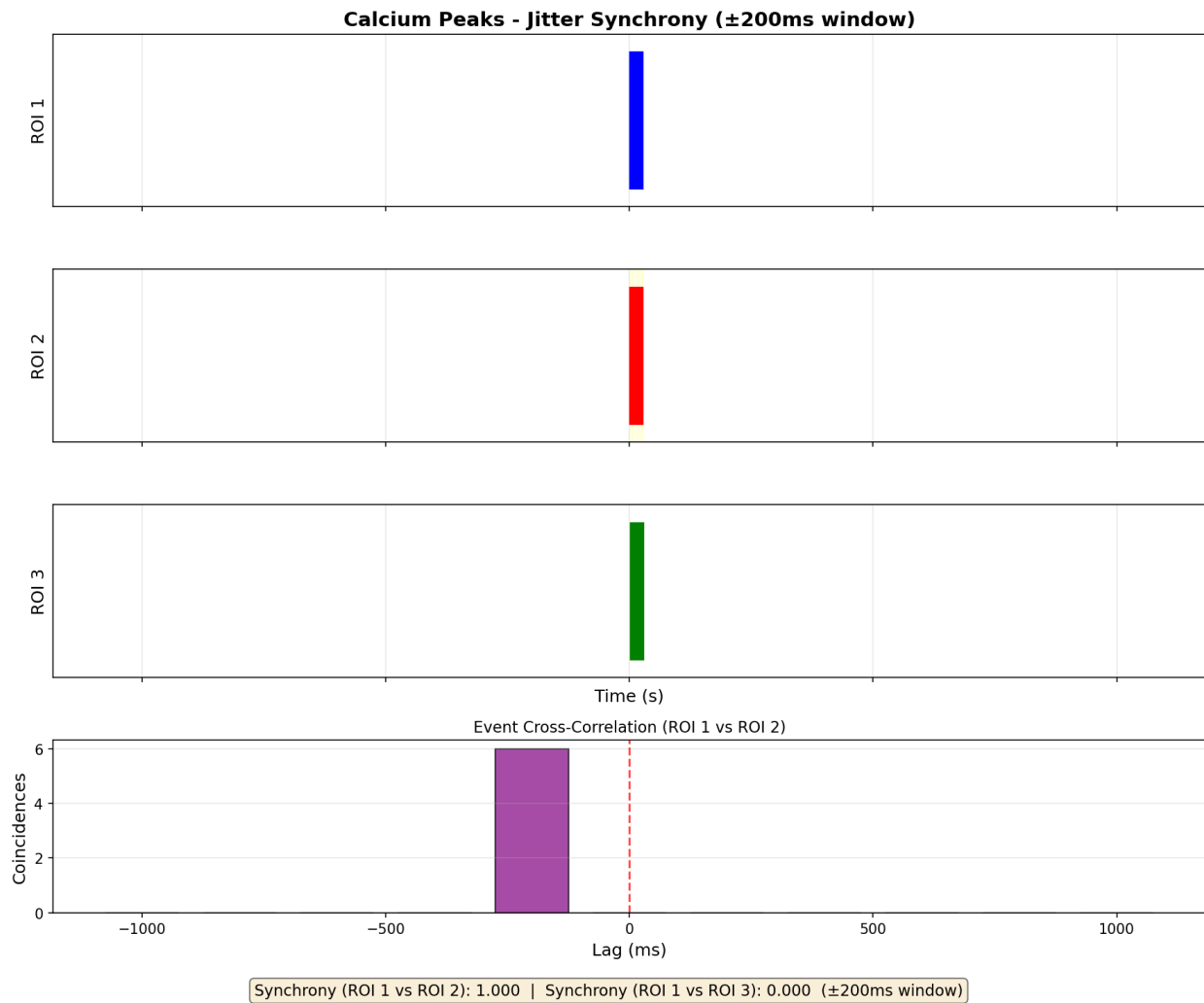


Figure 3: Calcium Peaks Synchrony

Yellow bands show the jitter window around ROI 1 peaks. When ROI 2 has a peak within this

window, the events are considered synchronous.

The **cross-correlogram** (bottom panel) shows how many coincident events occur at different time lags. A peak near zero lag indicates strong synchrony.

Interpretation

- **High synchrony (>0.7)**: ROIs frequently activate together within the time window
- **Moderate synchrony (0.3-0.7)**: Some co-activation, but also independent events
- **Low synchrony (<0.3)**: ROIs activate independently

When to use

Use for: - Detecting functional coupling between neurons - Identifying co-active neural assemblies
- Measuring ensemble synchronization (with appropriate jitter tolerance)

Avoid: - **Zero-lag Pearson correlation** on peak times (discrete events, not continuous signals) - Assuming millisecond-precision timing - Small jitter windows (<50 ms) — not supported by calcium imaging resolution

4. Inferred Spikes (Thresholded OASIS $s(t)$)

Inferred spikes are **discrete spike-like events** with partially restored temporal structure. Here, both correlation **and** synchrony metrics are appropriate.

Recommended Metrics

A. Zero-lag Pearson Correlation (on spike trains)

What it measures

Whether two neurons tend to be **active in the same time bins**: - Ensemble coactivity - Population coupling - Shared participation in network states

B. Max-Lag Cross-Correlogram (CCG)

What it measures

- **Connection strength**: Height of the CCG peak
- **Preferred temporal lag**: Position of the peak
- **Directionality**: Which neuron fires first (positive vs. negative lag)

C. Jitter Synchrony ($\pm$ window)

What it measures

Precise synchrony between spike trains within a narrow time window (e.g., ± 50 ms): - More robust than plain CCG - Returns a normalized synchrony score [0, 1] - Best measure of precise co-firing

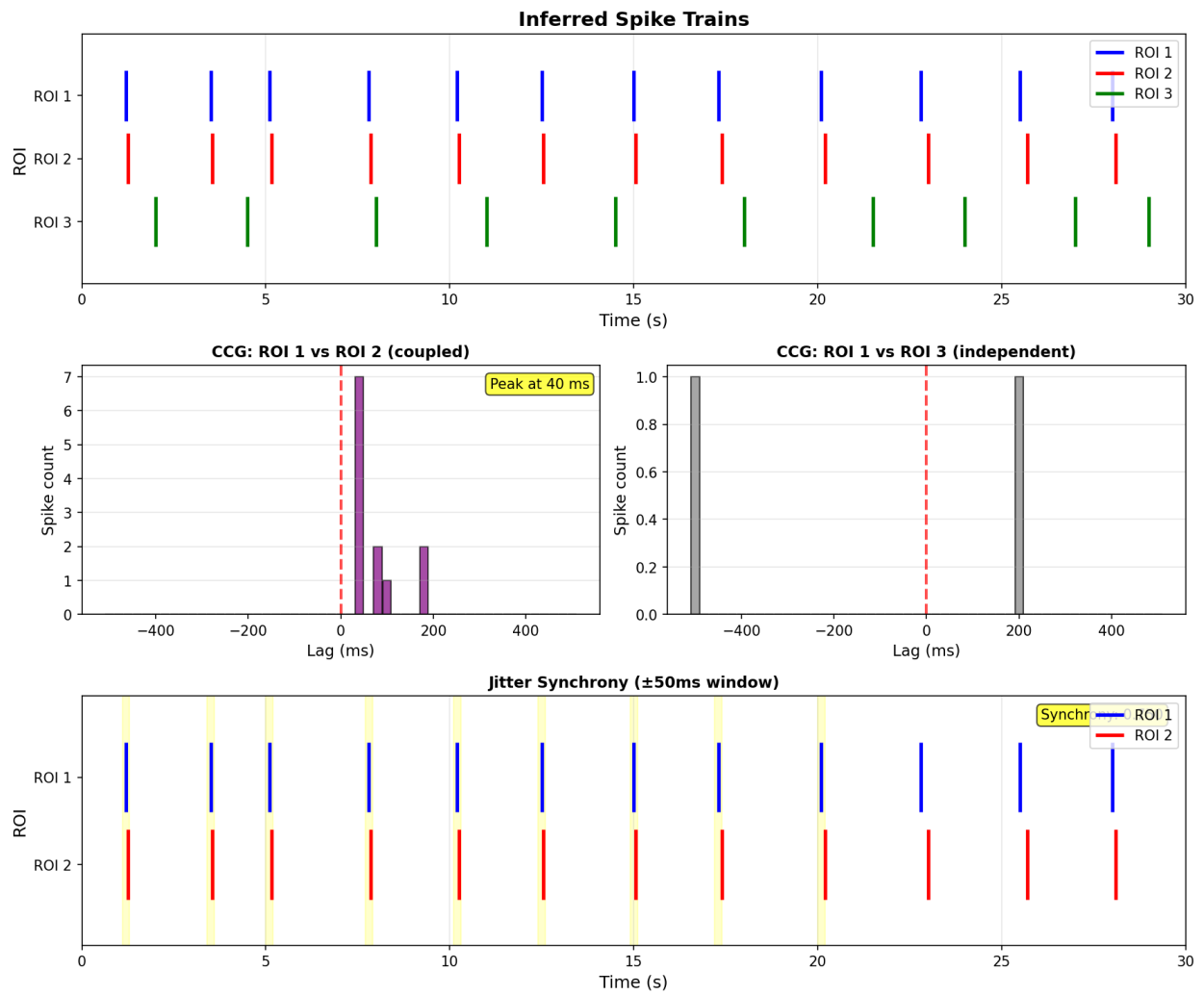


Figure 4: Spike Trains CCG and Synchrony

Visual Example

Top panel: Raster plot showing spike times for three ROIs.

Middle panels: Cross-correlograms (CCG) showing the number of spike coincidences at different time lags. A peak at positive lag indicates ROI 1 tends to fire before ROI 2.

Bottom panel: Jitter synchrony with a ± 50 ms window. Yellow bands highlight synchronous spike pairs.

Interpretation

Pearson Correlation

- **High ($r > 0.5$):** Neurons frequently fire in the same time bins
- **Low ($r \approx 0$):** Independent firing patterns

CCG

- **Peak at lag 0:** Synchronous firing
- **Peak at positive lag (+X ms):** ROI 1 leads ROI 2 by X ms
- **Peak at negative lag (-X ms):** ROI 2 leads ROI 1 by X ms
- **Flat CCG:** No temporal relationship

Jitter Synchrony

- **Score > 0.7 :** Strong synchrony
- **Score $0.3-0.7$:** Moderate synchrony
- **Score < 0.3 :** Weak/no synchrony

When to use

Use for: - Detailed analysis of spike timing relationships - Detecting monosynaptic connections (narrow CCG peaks) - Polysynaptic pathways (broader CCG peaks) - Precise synchrony analysis

Avoid: - Overinterpreting millisecond-precision (calcium imaging has limits) - Ignoring the jitter inherent in spike inference

Summary Table

Signal Type	Recommended Correlation	Recommended Synchrony	Avoid
$\Delta F/F$ Traces	Zero-lag Pearson	None (inappropriate)	Jitter synchrony, event metrics
Deconvolved C(t)	Zero-lag Pearson	None (inappropriate)	Timing-based synchrony
Calcium Peaks	Event Cross-Correlation (with lag)	Jitter synchrony ($\pm$ window)	Zero-lag Pearson correlation
Inferred Spikes	Pearson, Max-lag CCG	Jitter synchrony, CCG	Assuming perfect spike precision

Key Principles

1. **Match the metric to the signal type:** Continuous traces need correlation; discrete events can use synchrony.
 2. **Respect temporal resolution:** Calcium imaging is inherently slow (~100-300 ms jitter). Don't overinterpret millisecond-level precision.
 3. **Use lag tolerance:** For event-based metrics (peaks, spikes), always allow for temporal jitter.
 4. **Correlation Causality:** High correlation shows co-fluctuation, not directionality. Use CCG or other causal inference methods for directionality.
 5. **Deconvolution helps but has limits:** $C(t)$ and $s(t)$ improve temporal precision but don't fully restore spike timing.
-

Computational Notes

- **Zero-lag Pearson correlation:** Fast, works on any length traces

```
from scipy.stats import pearsonr
r, p = pearsonr(trace1, trace2)
```

- **Event cross-correlation:** Compute coincidences at different lags

```
# Pseudo-code
for lag in range(-max_lag, max_lag):
    coincidences[lag] = count_events_within_window(events1, events2, lag, window)
```

- **Jitter synchrony:** Fraction of events in one train that have a partner in the other within $\pm$ window

```
# Pseudo-code
sync_count = 0
for event1 in events1:
    if any(abs(event1 - event2) <= window for event2 in events2):
        sync_count += 1
synchrony = sync_count / len(events1)
```

- **CCG for spike trains:** Histogram of inter-spike intervals across neurons

```
# Available in many packages, e.g., elephant, NEO
```